

3-1. Outline

The LH5810/LH5811 is a single chip CMOS static LSI that features the following functions:

- (1) two pairs of 8-bit bidirectional port
- (2) one pair of 8-bit output port
- (3) two interrupt request inputs (one of them port input)
- (4) one interrupt request output
- (5) CPU wait control
- (6) serial data transfer control

3-2. Functions

- ① Ports PA0~7 and PB0~7 can be programmed of their data flow direction in bit unit. Also, it can be accessed as one location of the memory, as seen from the CPU.
- ② Latch clock $P\Phi$ can be directly given from the external source through output ports PC0~7. Also, it can be accessed as one location of the memory, as seen from the CPU.
- ③ As there are two interrupt request inputs of IRQ and PB7, interrupt request can be issued to the CPU at the rising edge of the input when the corresponding bit of the MSK register is "1". PB7 must be in the input mode before using PB7 for the interrupt input.
- ④ Since there is the CPU wait control circuit, two memory enable signals can be output to the memory of slow access time. Besides, it has two inputs of wait conditions. Up to 8 varieties of access time can be programmed.
- ⑤ It has the following functions to handle serial data transfer.

A. Serial data transmit

Serial data transfer takes place in a format of a start bit, 8 bits of data, and two stop bits. Transmission clock is selectable by means of internal and external clock select program, as well as the clock rate ($1/1$, $1/2$, $1/128$, $1/256$, $1/512$, $1/1024$, $1/2048$, and $1/4096$ of the basic clock).

B. Serial data receive

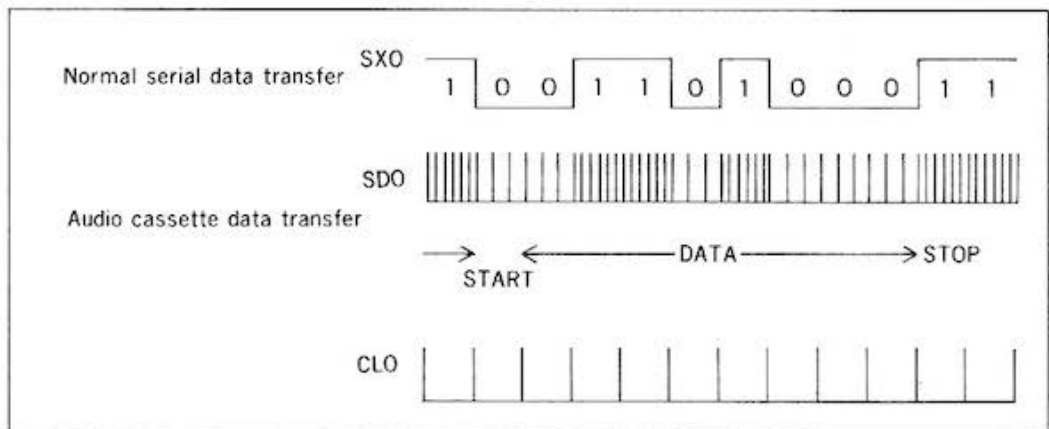
When the start bit is received in an idle state, the serial data following to it is received. After receiving the data comprised of 8 bits, it is then stored in the internal register and the interrupt request flag is set active. Receive clock is furnished from the external source which becomes the receiving clock by itself. It must be in synchronization with the serial data input.

C. Pulse waveform

It is possible to have continuous output of pulse waveform. Frequency is programmable to eight kinds of $1/1$, $1/2$, $1/128$, $1/256$, $1/512$, $1/1024$, $1/2048$, and $1/4096$ of the basic clock.

D. Data transfer to the audio cassette tape

Format of data transferred to the audio cassette tape consists of a start bit, 8 bits of data, and two stop bits, with the modulation signal generated from the SDO output.



Assume now that normal serial data output is to be SDO, the modulation clock to the data 1 to be FX, the modulation clock to the data 0 to be FY, and audio cassette tape data output to be SDO, then the following equation comprises:

$$SDO = SDO \cdot FX + \overline{SDO} \cdot FY$$

Whereas, FX and FY can be set independently to 1/64, 1/128, 1/256, 1/512 or 1/1024 of the basic clock by means of programming.

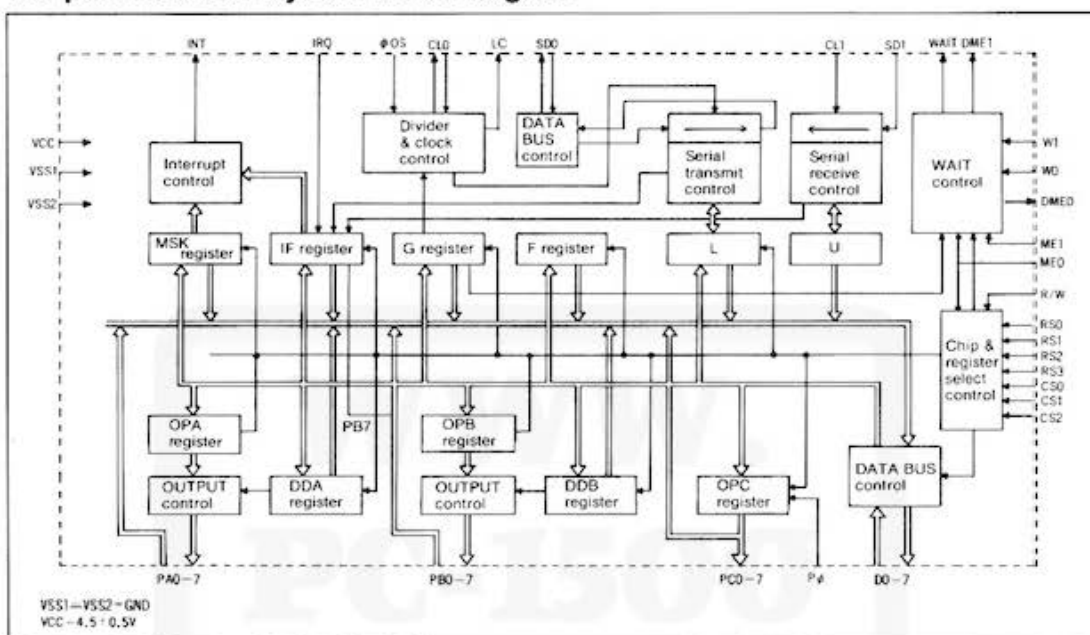
The serial transmit clock CLO can be programmed as discussed in Item A.

3-3. Internal structure

3-3-1. Block diagram

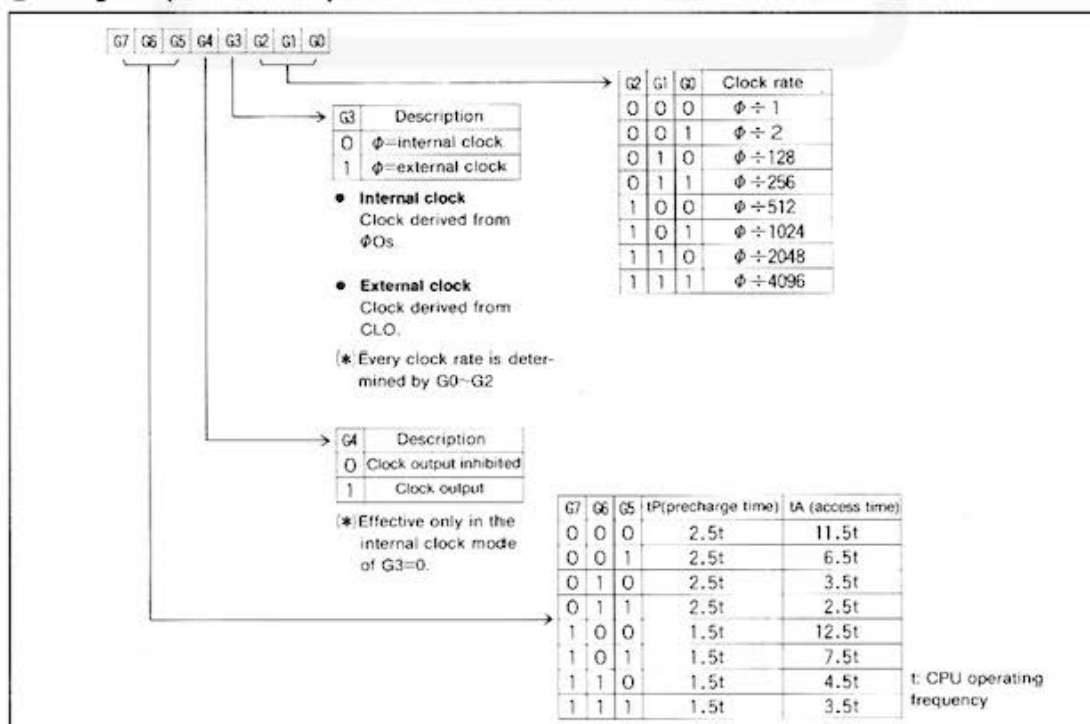
The I/O port controller consists of nine internal registers, wait controller, serial controller, and interrupt controller, and each of internal registers can be accessed as one location of the memory as seen from the CPU.

I/O port controller system block diagram



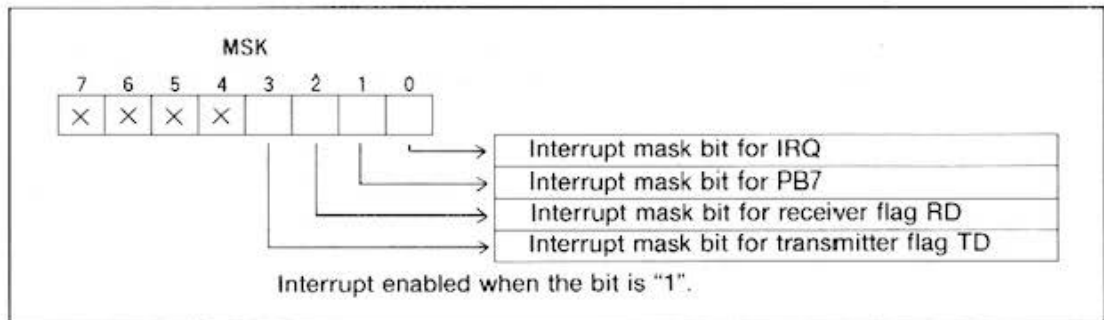
3-3-2. Internal registers

① G register (RS3~0=1001)



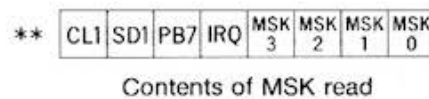
The G register can read/write data when register is selected (1001).

② MSK register

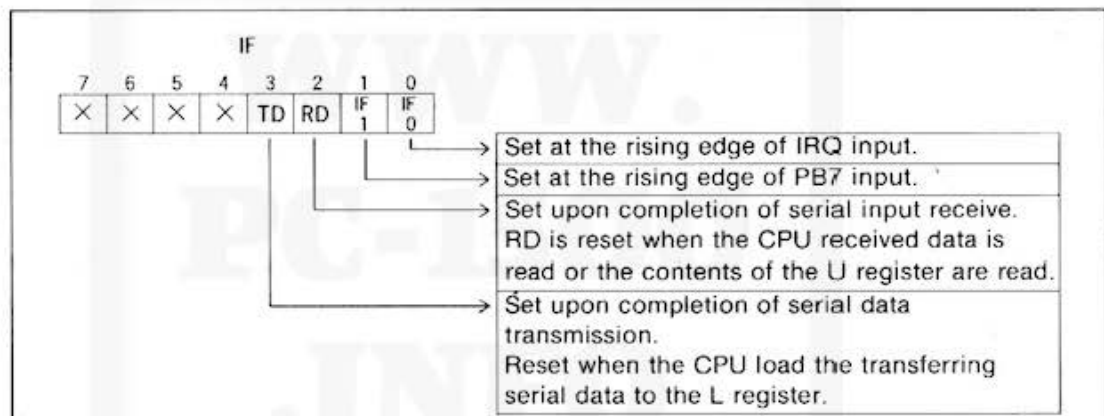


The MSK register can read/write data when register is selected (1010).

NOTE: When the contents of the MSK register are read, the contents of CL1, SD1, PB7, and IRQ are stored in high order digit positions.



③ IF register

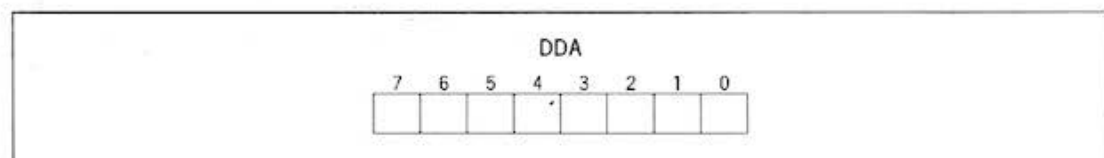


IF0 and IF1 can read/write data when register is selected (1011).

RD and TD are dedicated to read only.

NOTE: During receive of serial data, RD is reset. Term "serial data receive" means the period during which an 8-bit data is in reception, with the start bit excluded.

④ DDA register



The register used to determine the direction of the port PA.

i-th bit of the DDA register

When 0	PA _i is in the input mode.
When 1	PA _i is in the output mode and outputs the contents of OPA _i .

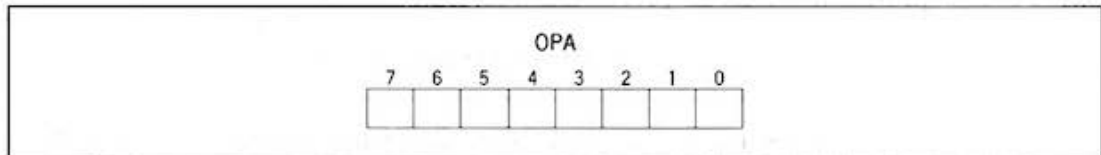
DDA can read/write when the register is selected (1100).

⑤ DDB register

The register used to determine the direction of the port PB.

Selection of the input/output port is the same as in the DDA register.

DDB can read/write data when the register is selected (1101).

⑥ OPA register

The OPA register is the buffer for input and output of data to the port PA.

In the case of output, DDA_i must be set to "1" (output mode) and the register must be selected (1110), then the data on the bus line is loaded into OPA_i, to be output on PA_i.

In the case of input, DDA_i must be set to "0" (input mode) which prohibits output from OPA_i, then the contents of PA_i are loaded into OPA_i and the data is sent on the bus line.

⑦ OPB register

The OPB register is the buffer for input and output of data to the port PB.

It has the same function as the OPA register.

OPB can read/write data when the register is selected (1111).

⑧ OPC register

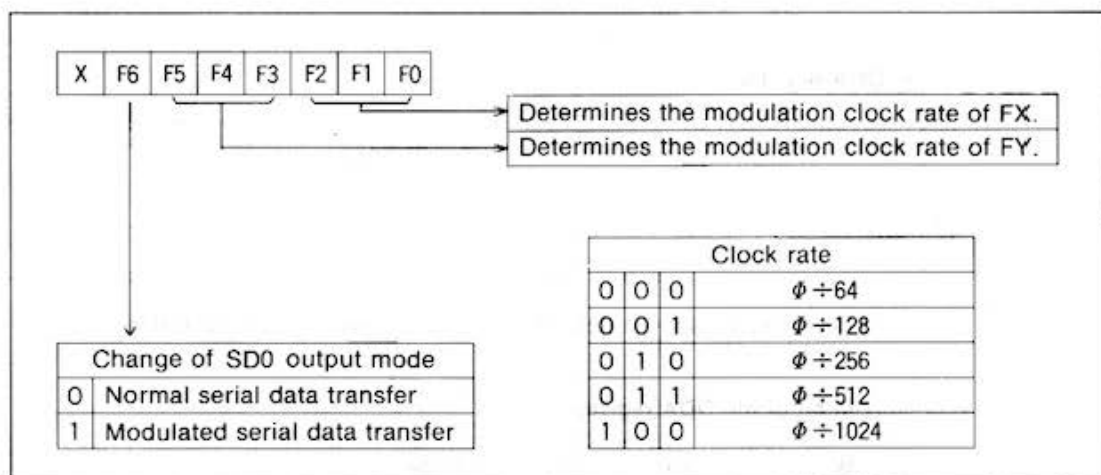
The OPC register can read/write data when the register is selected (1000).

Also, the contents of the data bus can be latched to the OPC register at the falling edge of the external input clock PΦ.

⑨ F register

The F register can read/write data when the register is selected (0111).

F0-2 determines the clock rate of the modulation clock FX and F3-5 the modulation clock of FY.



3-3-3. Pin description

(*): See Pad layout and structure in 3-5-2.

No.	Signal name	Function	No.	Signal name	Function
1	PA1	Port input/output	31	RS2	Register select input
2	PA2	Port input/output	32	RS3	Register select input
3	PA3	Port input/output	33	R/W	Read/write input
4	PA4	Port input/output	34	ME0	Memory enable input
5	PA5	Port input/output	35	ME1	Memory enable input
6	PA6	Port input/output	36	W0	Wait condition input
7	PA7	Port input/output	37	W1	Wait condition input
8	GND	Power source	38	GND	Power source
9	PB0	Port input/output	39	VCC	Power source
10	PB1	Port input/output	40	DME0	Memory enable output
11	PB2	Port input/output	41	DME1	Memory enable output
12	PB3	Port input/output	42	WAIT	Wait output
13	PB4	Port input/output	43	INT	Interrupt output
14	PB5	Port input/output	44	RESET	Initialize output
15	PB6	Port input/output	45	IRQ	Interrupt input
16	PB7	Port input/output, Interrupt input	46	ϕ OS	Basic clock input
17	P ϕ	Port PC latch clock	47	CL1	Serial receive clock input
18	PC0	Port output	48	SD1	Serial receive input
19	PC1	Port output	49	LC	Not used
20	PC2	Port output	50	CL0	Serial rec/trn clock input/output
21	PC3	Port output	51	SD0	Serial rec/trn input/output
22	PC4	Port output	52	D0	Data bus input/output
23	PC5	Port output	53	D1	Data bus input/output
24	PC6	Port output	54	D2	Data bus input/output
25	PC7	Port output	55	D3	Data bus input/output
26	CS0	Chip select input	56	D4	Data bus input/output
27	CS1	Chip select input	57	D5	Data bus input/output
28	$\overline{\text{CS2}}$	Chip select input	58	D6	Data bus input/output
29	RS0	Register select input	59	D7	Data bus input/output
30	RS1	Register select input	60	PA0	Port input/output

3-4. Functions

3-4-1. Operation

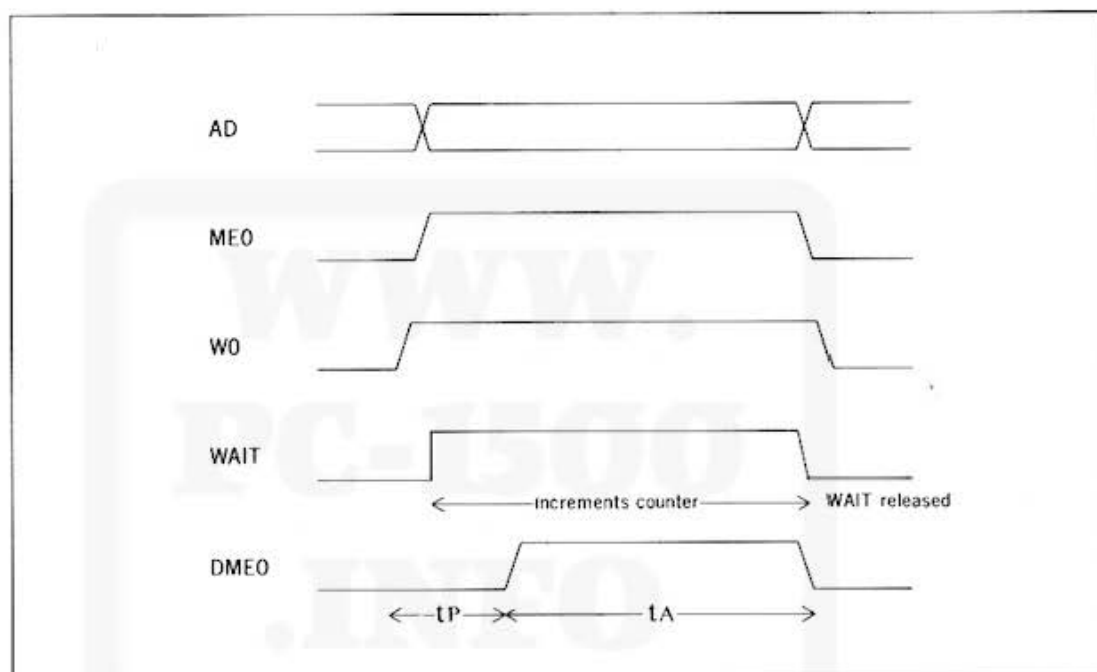
I/O port controller operation

CS2	CS1	CS0	RS3	RS2	RS1	RS0	R/W	Operation
0	1	1	0	1	0	0	0	Resets the divider internally contained in the I/O port chip.
			0	1	0	1	1	Reads the contents of the U register and sends them on the data bus. At the same time, the receive flag RD is reset.
			0	1	1	0	0	The contents of the data bus are converted into a serial data signal of start/data/stop structure. At the same time, the transmit flag TD is reset.
			0	1	1	1	0	The contents of the data bus are stored in the F register.
			1	0	0	0	0	The contents of the data bus are stored in the OPC register.
							1	The contents of the OPC register are sent on the data bus.
			1	0	0	1	0	The contents of the data bus are stored in the G register.
							1	The contents of the G register are sent on the data bus.
			1	0	1	0	0	The contents of the data bus are stored in the MSK register.
							1	The contents of the MSK register are sent on the data bus.
			1	0	1	1	0	The contents of the data bus are stored in the IF register.
							1	The contents of the IF register are sent on the data bus.
			1	1	0	0	0	The contents of the data bus are stored in the DDA register.
							1	The contents of the DDA register are sent on the data bus.
			1	1	0	1	0	The contents of the data bus are stored in the DDB register.
							1	The contents of the DDB register are sent on the data bus.
			1	1	1	0	0	The contents of the data bus are stored in the OPA register.
							1	The contents of the PA _i are sent on the data bus.
			1	1	1	1	0	The contents of the data bus are stored in the OPB register.
							1	The contents of the PB _i are sent on the data bus.

3-4-2. Wait control

① Function

Wait is a function applied in accessing a slow action memory. It makes the CPU operation temporarily halted until a complete access is done to the slow action memory. When the CPU makes access to the address where slow action memory is assigned, the condition "1" is entered to the wait condition inputs W0 and W1. W0 is the wait condition input which is applied to place wait to the memory area controlled by ME0, while W1 is the wait condition for the memory area controlled by ME1. When the wait condition is established, that is, when " $W0 \cdot ME0 + W1 \cdot ME1 = 1$ " is met, the WAIT signal is issued to the CPU. As the wait control counter is provided internally, it releases WAIT to the CPU when it is counted to the value set by the program, then the CPU proceeds to execute a next machine cycle.



Since a slow access memory usually consists of dynamic logic, it needs a precharge time (t_P). Thus, the I/O port controller issues the memory enable signals DME0 and DME1 to the dynamic memory, including the precharge time. DME0 is for ME0 and DME1 is for ME1.

When the CPU accesses the I/O port controller, the signal WAIT is issued without the wait condition in W0 and W1.

② Wait time

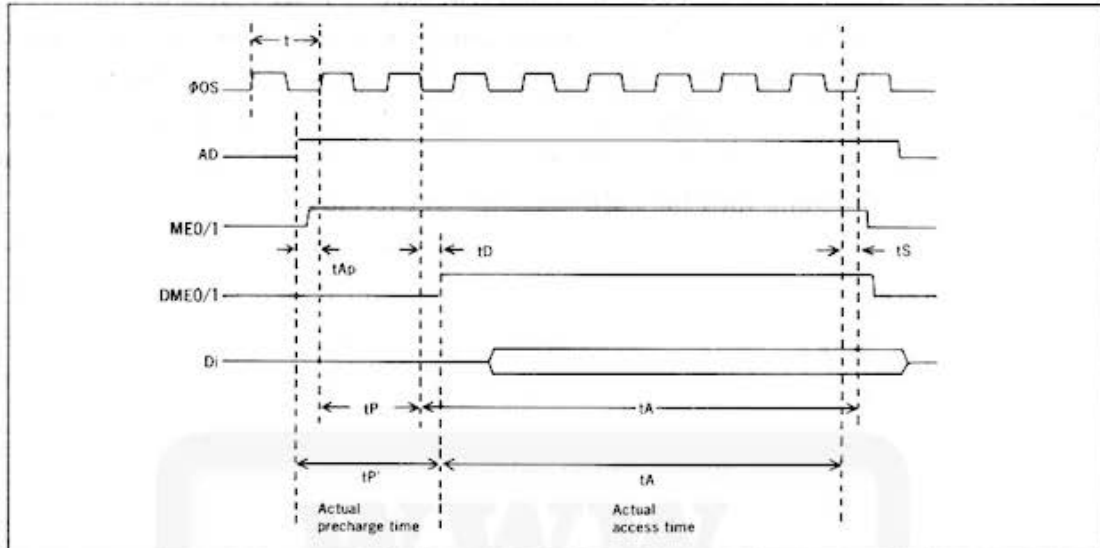
Wait time can be programmed by means of bit positions of G5, 6 and 7 of the G register. Relation of G5, 6, 7 with t_A and t_P refer to page 70.

③ Wait time of I/O port controller itself

When the CPU accesses the I/O port controller, wait of a single CPU machine cycle is automatically applied.

If the W0 and W1 is in the wait state when the CPU accessed the I/O port controller, wait ends in one cycle.

NOTE: When there is no wait condition in W0 and W1, the same waveform as ME0 and ME1 are sent on DME0 and DME1.



tP' and tA' for actual ROM is as follows:

$$tP' = tAP + tP + tD$$

$$tA' = tA - tD - tS$$

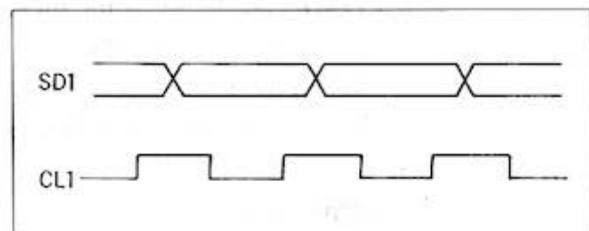
$$tS = 200nS$$

Since tAP and tD differ depending on the peripheral circuit configuration, they should be computed on the basis of load capacitance.

3-4-3. Serial data input

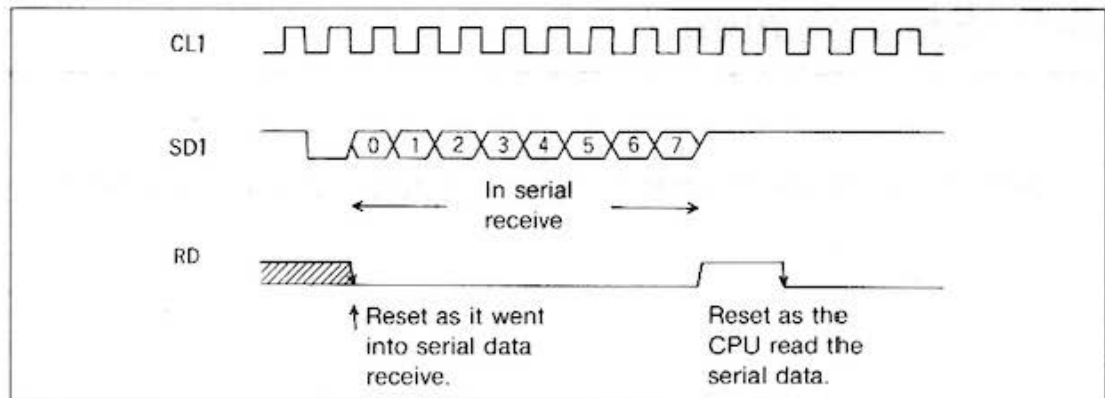
SDI is a serial data transfer input and CLI serial data transfer clock.

The I/O port controller reads the input data at the rising edge of CLI. Serial data goes into the receiving mode when it changes from an idle state ($SDI=1$) to low state and reads data from a next clock.



When the 8-bit data is received, it sets the receive and flag RD active. If the mask bit is on at this point, interrupt request is issued to the CPU. RD will be reset upon reading the 8-bit data.

RD will be reset in a course of serial data receiving, which is the period that the 8-bit data is being received, without including the start bit.



3-4-4. Reset

When the RESET line is kept in "1" at least for three ϕ OS clock cycles, it causes internal reset, by which all internal registers are cleared to "0" and ports PA and PB go into the input mode. The divider, however, will not be reset.

Terminal states after the reset are as shown in Table below.

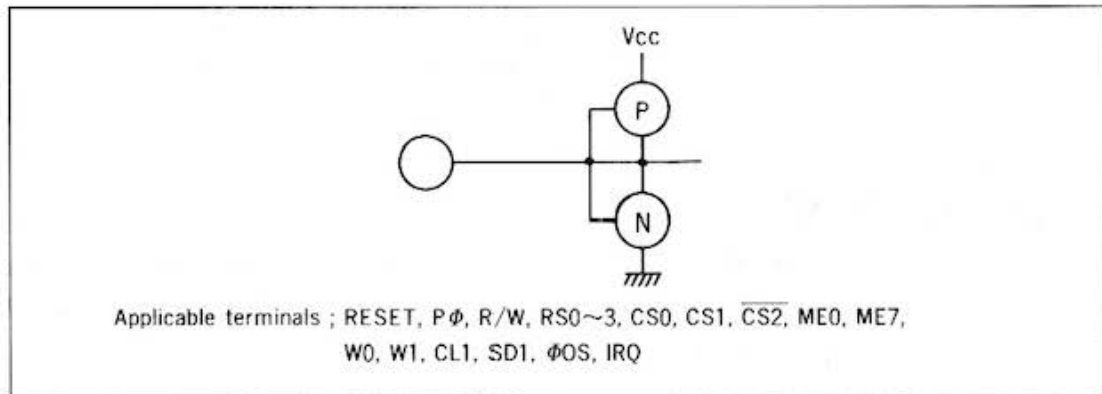
Pin name	In/Out	State after reset
D0~D7	In/Out	High impedance
PA0~PA7	"	"
PB0~PB7	"	"
PC0~PC7	Out	Low level
WAIT	"	(NOTE)
DME0	"	"
DME1	"	"
INT	"	Low level
SD0	In/Out	High level
CL0	"	Low level

NOTE: WAIT, DME0, and DME1 are output dependent of W0, ME0, W1, and ME1 and do not have any connection with reset.

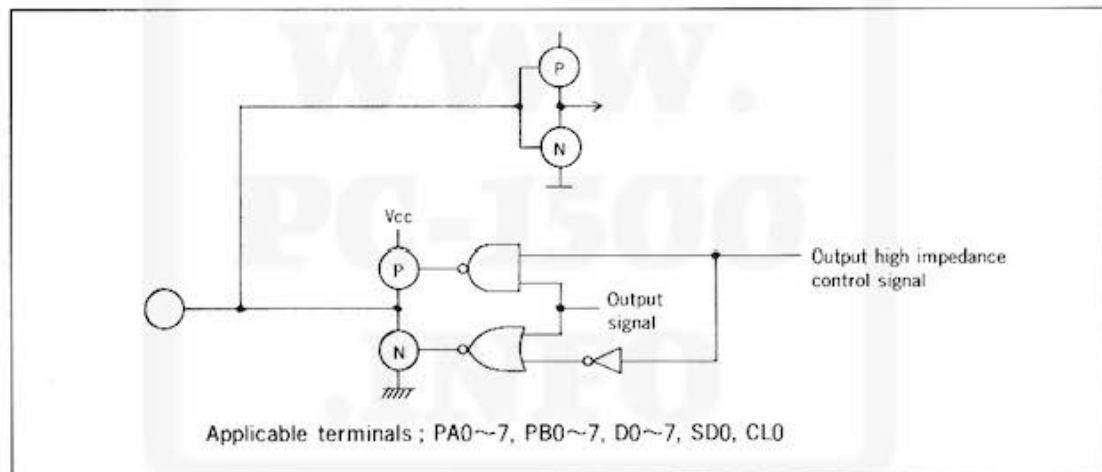
3-5. Specification

3-5-1. I/O port controller input/output circuits

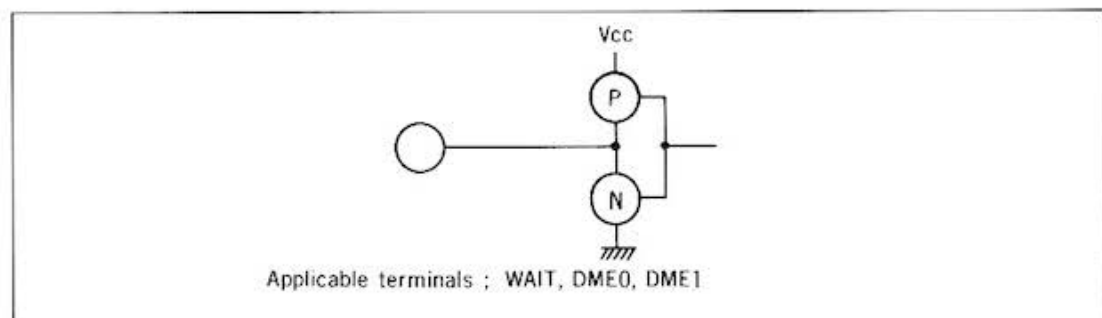
① Input



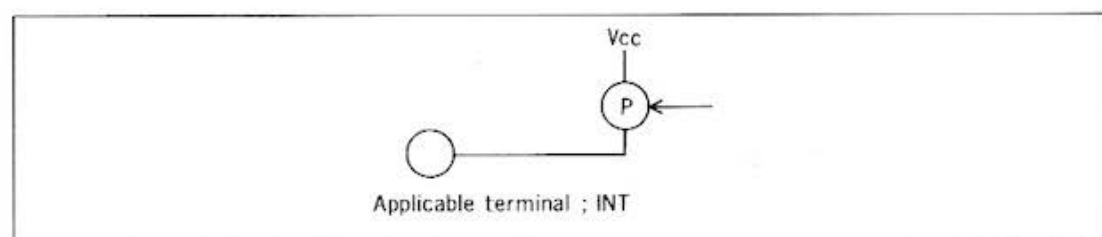
② Input/output



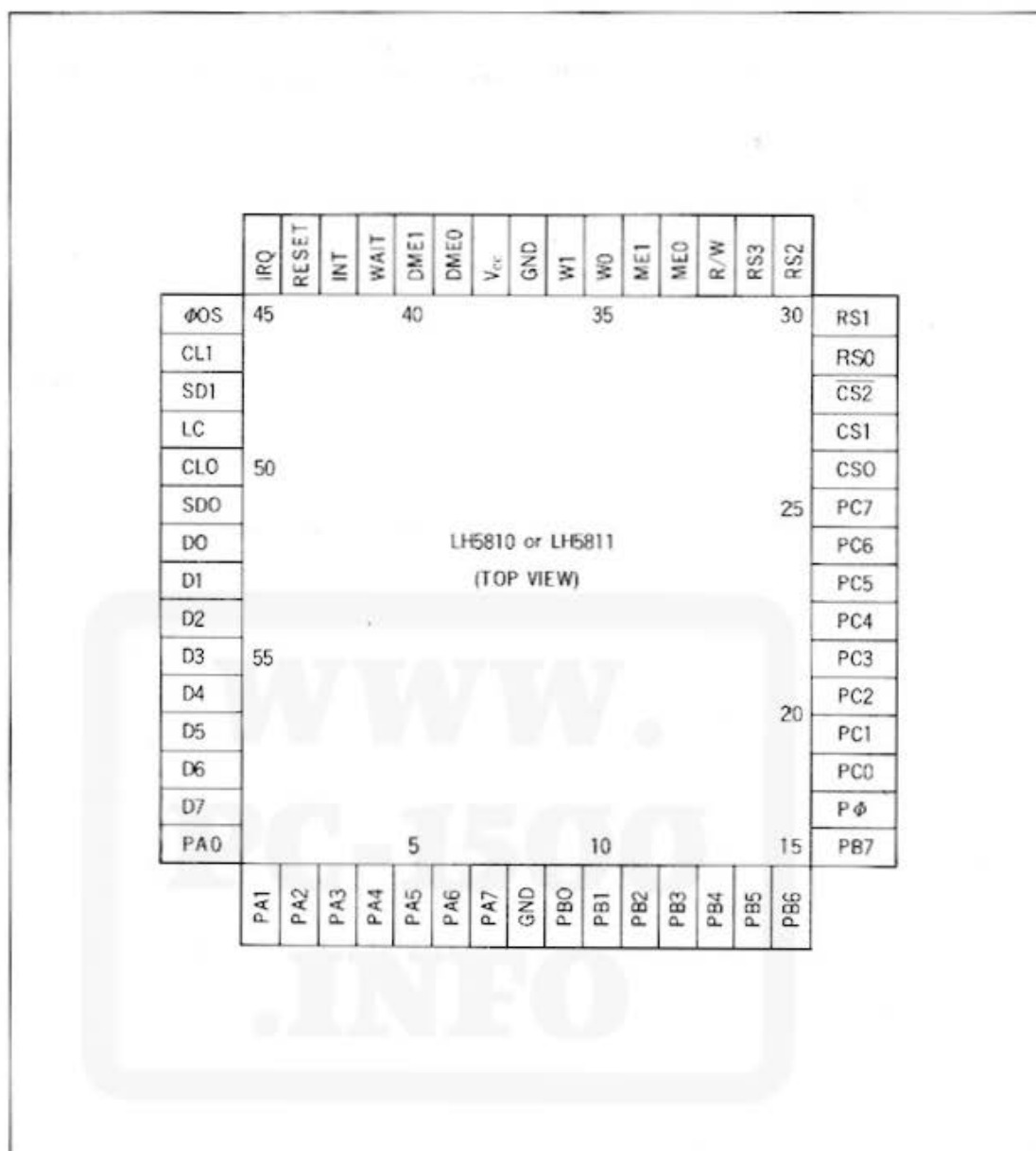
③ Output



④ Output



3-5-2. Pad layout and structure



3-5-3. Electrical characteristics

① Absolute maximum ratings

Parameter	Symbol	Limits	Unit
Input apply voltage (*1)	V_{in1}	-0.3 to 6.5	V
Input apply voltage (*2)	V_{in2}	-0.3 to V_{cc} +0.3	V
Operating temperature	T_{opr}	-5 to +55	°C
Storage temperature	T_{stg}	-55 to +150	°C

*1: Applicable to V_{cc} and with respect to GND.

*2: Applicable to other than V_{cc} , GND, and with respect to GND.

② Operating condition

Parameter	Symbol	Limits	Unit
Supply voltage	V_{cc}	4.0 to 5.0	V

③ Electrical characteristics

③ - 1 - DC characteristics

$T_a = -5 \sim 55^\circ\text{C}$
 $V_{CC} = 4.0 \sim 5.0\text{V}$

Parameter	Symbol	Limits			Unit	Test condition	Note
		Min	Typ	Max			
Input voltage	V_{IL}	0		0.8	V		1
	V_{IH}	2.4		V_{CC}	V		
Output voltage 1	V_{OL1}			0.4	V	$I_{OL} = 1.6\text{mA}$	2
	V_{OH1}	2.4			V	$I_{OH} = 0.4\text{mA}$	
Output voltage 2	V_{OL2}	1.5			V	$I_{OH} = 1.5\text{mA}$	3
Output leakage current	$ I_{LO} $			1.0	μA		4
Input current	$ I_{II} $			1.0	μA	$V_{IN} = 0\text{V}$ or V_{CC}	5
Normal power dissipation	I_{CCA}		0.4	0.9	mA	$\phi OS = 2\text{MHz}$	6
	I_{CCB}			5	μA	$\phi OS = 0\text{V}$	7

NOTES: For Note number, refer to the list below.

Note No.	Applicable terminals
1	PA0~7, PB0~7, D0~7, R/W, RS0~3, CS0, CS1, $\overline{CS2}$, ME0, ME1, W0, W1, SD1, CL1, ϕOS , SD0, CL0, IRQ, RESET.
2	PA0~7, PB0~7, PC0~7, D0~7, DME0, DME1, WAIT, CL0, INT
3	PA0~7, CL0, SD0.
4	All input terminals.
5	All output terminals.
6	The term "normal" applies to the reset state that the I/O port controller is not selected and the 2MHz clock should be supplied through the ϕOS pin. The following input terminals should be connected to 0V except $\overline{CS2}$. PA0~7, PB0~7, D0~7, CS0, CS1, $\overline{CS2} = V_{CC}$, R/W, RS0~3, $P\phi$, ME0, ME1, W0, W1, CL1, SD1, IRQ.
7	Applied to the state that no clock is supplied to the ϕOS terminal in the normal condition.

③ - 2 - AC characteristics

(Ta = -5 to 55°C, V_{CC} = 4.0 to 5.0V)

Parameter	Symbol	Limits			Unit	Test condition
		Min	Typ	Max		
CS0, CS1, $\overline{CS2}$, RS0~3	t _D			230	ns	
	t _F	80			ns	
ME0, ME1	t _D			240	ns	
	t _F	80		175	ns	
	t _{WM}	250			ns	
WAIT	t _{WS}	150			ns	C _L = 50pF
	t _{WF}	50		200	ns	
D0~7 (input mode)	t _D			500	ns	V _{OH} = 0.8V _{CC}
	t _F	130			ns	
	t _{ZD}	100			ns	
	t _{ZF}			300	ns	
D0~7 (output mode)	t _S	200			ns	
	t _F	50			ns	
	t _{ZM}	150			ns	
	t _{ZF}			100	ns	
R/W	t _D	-30		50	ns	V _{OH} = 0.8V _{CC}
	t _F			150	ns	
ΦOS	t _{RC}	500			ns	0.1~0.9V _{CC}
	t _r			150	ns	
	t _f			40	ns	
PA0~7, PB0~7 (output mode)	t _{OR}			1	μs	C _L = 100pF
PC0~7 (register select)	t _{OR}			1	μs	C _L = 100pF
PΦ	t _D			350	ns	V _{OH} = 0.8V _{CC}
	t _F			100	ns	
PC0~7 (PΦ latch)	t _{OP}			1	μs	C _L = 100pF
SD0 (input mode) SD1	t _S	100			ns	
	t _H	50			ns	
SD0 (output mode)	t _D			100	ns	C _L = 100pF
DME0 DME1	t _D			120	ns	C _L = 50pF
	t _F			120	ns	
SD0	t _Z			200	ns	C _L = 100pF
CL0	t _r			100	ns	0.1~0.9V _{CC} C _L = 50pF
	t _f			100	ns	0.9~0.1V _{CC} C _L = 50pF

(*): Unless otherwise specified, the criterion shall be V_{OH} = 2.0V, V_{OL} = 0.4V

③ - 3 - Timings

